

FEM: from an edited CAD solid to a checked response

1. Define the scenario

Solid + material
Fixed band + loaded band

2. Assemble and solve

$Kq = f$
Linear tetrahedral elasticity

3. Check the solution

Forces, energy, mesh change
Compare edited vs. target

Blue: fixed nodes Red arrow: resultant load direction All surfaces are undeformed.

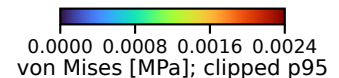
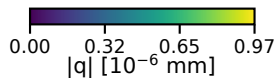
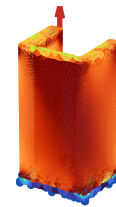
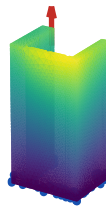
Mounting-angle width edit | actual final 3B prediction

33,871 nodes; 142,334 tets | $C = 7.461\text{e-}07 \text{ N mm}$ | $\max |q| = 9.750\text{e-}07 \text{ mm}$

Tetrahedral surface mesh

Displacement magnitude

Diagnostic stress field



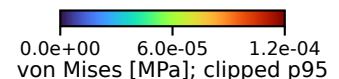
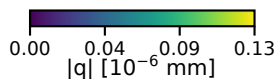
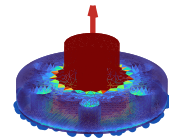
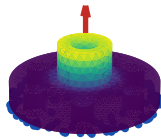
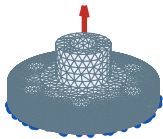
Flange: 4-to-6 mounting-hole edit | actual final 3B prediction

15,035 nodes; 59,326 tets | $C = 1.214\text{e-}07 \text{ N mm}$ | $\max |q| = 1.331\text{e-}07 \text{ mm}$

Tetrahedral surface mesh

Displacement magnitude

Diagnostic stress field



Worked-example interpretation

The same solid supplies the mesh and both field plots. A color change is a solver response, not proof that the edit is correct: independent solid and target-response checks are still required.

Assumed $E = 210,000 \text{ MPa}$, $\nu = 0.3$; static 1 N test. Stress convergence / operating safety are not established. Surface edges display the tetrahedral boundary, not every interior element.